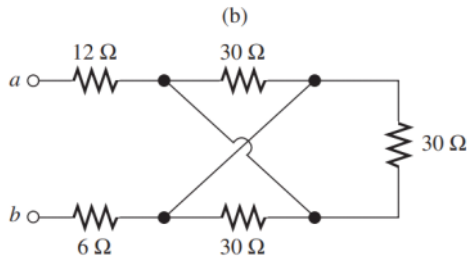
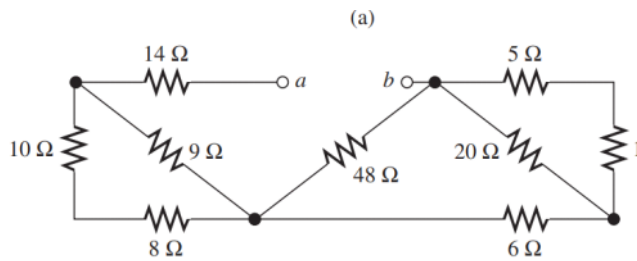
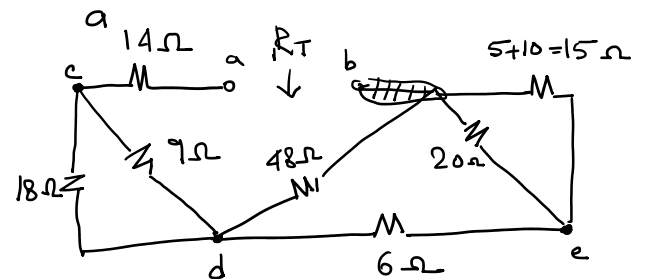
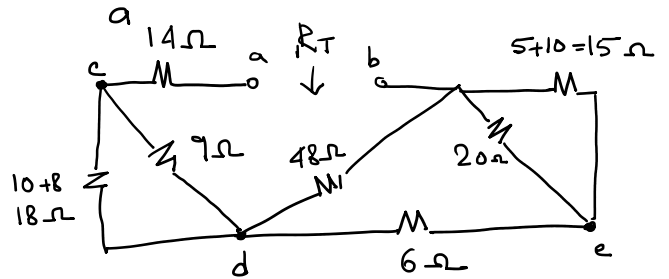


Equivalent CKT

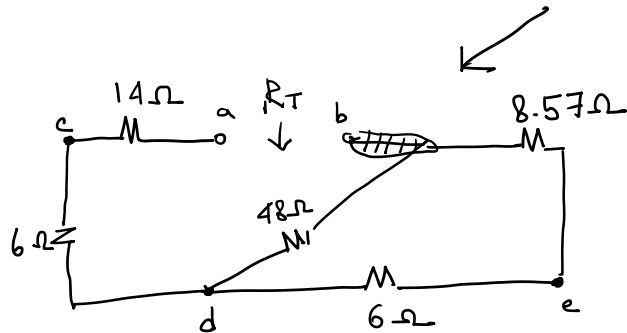
Wednesday, November 13, 2024 8:19 PM



(c)

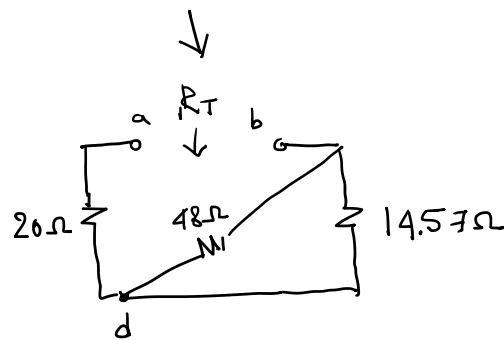


$$c \& d \rightarrow 18\Omega \parallel 9\Omega = \frac{18 \times 9}{27} = 6\Omega, \quad b \& e \rightarrow 20\Omega \parallel 15\Omega = \frac{20 \times 15}{35} = 8.57\Omega$$

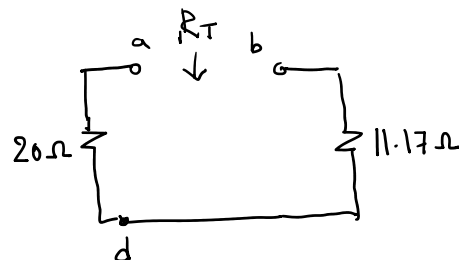


$$a \rightarrow c \rightarrow d: 14\Omega + 6\Omega = 20\Omega \text{ [Series]}$$

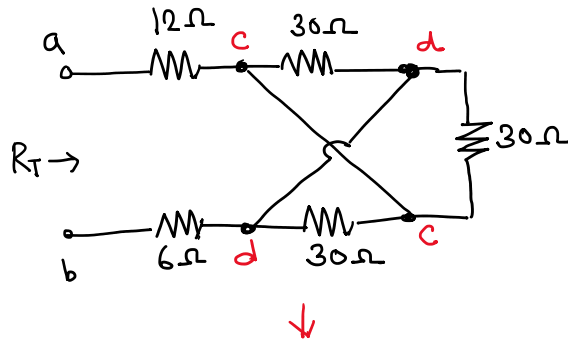
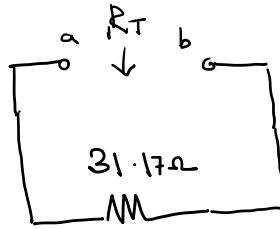
$$b \rightarrow e \rightarrow d: 8.57\Omega + 6\Omega = 14.57\Omega \text{ ["]}$$



$$\text{between } b \& d \Rightarrow 48\Omega \parallel 14.57\Omega = \frac{48 \times 14.57}{62.57} = 11.17\Omega$$



$$R_T = 20\Omega + 11.17\Omega = 31.17\Omega$$

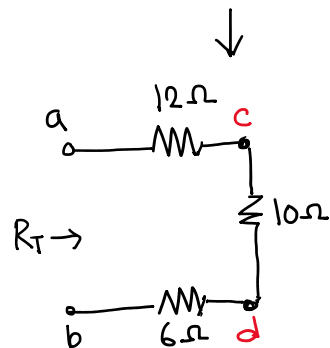
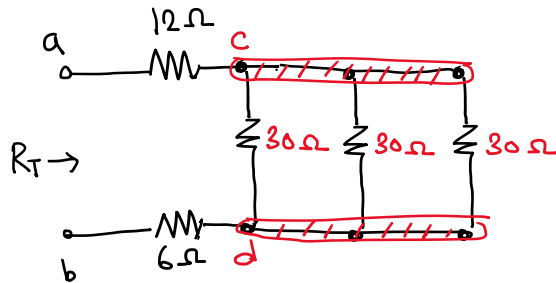


Between node c & d $\rightarrow 30\Omega$
 $c \& d \rightarrow 30\Omega$
 $c \& d \rightarrow 30\Omega$

Therefore, $30\Omega \parallel 30\Omega \parallel 30\Omega$

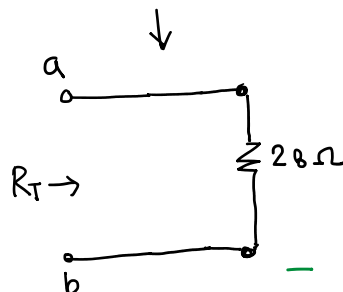
$$\frac{1}{R_1} = \frac{1}{30} + \frac{1}{30} + \frac{1}{30} = \frac{1}{10}$$

$$R_1 = 10\Omega$$



Now 12Ω , 10Ω & 6Ω are connected in series.

$$R_T = 12\Omega + 10\Omega + 6\Omega = 28\Omega$$



R_2

